

Zhanhe Shi

New York University

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Education

New York University

Master of Science in Information Systems

New York, NY, USA

Sep 2025 - May 2027

- **GPA: 4.0/4.0**
- **Courses:** Fundamental Algorithms, Deep Learning, Data Science & AI for Business

University of California, Berkeley

College of Engineering, Computer Science

Berkeley, CA, USA

Aug 2023 - May 2024

- **GPA: 3.8/4.0**
- **Courses:** Deep Neural Networks, Computer Vision and Computational Photography, Optimization Models in Engineering

Internship Experience

Poshmark, Inc.

AI/ML Engineering Intern

Redwood City, CA, USA

Jun 2026 - Aug 2026

- Engineered prompts and redesigned the **multi-step model-call pipeline** for SmartList, which drafts product listings from seller photos, now in A/B testing.
- Built an **evaluation pipeline** that scores listings field by field against seller-written ones to benchmark prompts, open-source models and APIs.
- Hosted open-weight vision-language judges on A100 GPUs via **vLLM**, raising **throughput per GPU by 150%** over the default serving configuration.
- Prototyped **automatic cropping** of seller photos using **SAM 3** segmentation and rule-based framing, outperforming the paid vendor in internal review.
- Built a **LangGraph** demo agent for offer negotiation that recommends a price from a **knowledge base** mined from the platform's negotiation records.

Hundsun Technologies Inc.

Intern, Junior Software Engineer

Hangzhou, Zhejiang, China

May 2024 - Jul 2024

- Constructed a **Document Layout Analysis benchmarking pipeline** on the Chinese CDLA dataset, evaluating 7 open-source and commercial models.
- **Fine-tuned state-of-the-art multimodal networks** (LayoutLMv3, VGT) and object detection models (YOLO), achieving a peak **mAP@50:95 of 88.4** with VGT.
- Accelerated inference via **ONNX and TensorRT**, reducing VGT backbone latency by **65%** on an NVIDIA A100 GPU.
- **Trained YOLOv7** for lightweight layout detection, minimizing memory footprint to 1GB and achieving a **103x speedup** (3.8ms).
- Spearheaded algorithmic research to deliver a zero-cost replacement, with the VGT **outperforming the commercial API by 64%** in title extraction.
- Delivered a **model selection matrix** to engineering, enabling production deployments that optimally balanced throughput and precision.

Research & Project Experience

Physics-Aware Representation Learning for Physical Systems

Co-Author, New York University

New York, NY, USA

Mar 2026 - May 2026

- Performed **self-supervised learning** on PolymathicAI active-matter simulations, evaluated by downstream physical-parameter regression.
- Designed **H-JEPA**, a **dual-pathway hierarchical encoder** that **cut test error by 70%** over the JEPA baseline at comparable **~4M parameters**.
- Implemented end-to-end in **PyTorch Lightning** with VICReg + EMA-target self-supervision and **bf16-mixed-precision** training on NVIDIA A100 GPUs.
- Delivered an open-source PyTorch codebase and a report, with **ablations** validating the hierarchical design and the choice of regularizer.

Multiple Human-Object Interaction Generation with Conditional Diffusion

Graduation Project, ShanghaiTech University

Shanghai, China

May 2025

- Proposed a diffusion framework with human pose and object movement, achieving a **high-fidelity generation** (FID of 8.95) on a subset of HOI-M³.
- Extracted object global features from **point clouds** by a PointNet++ encoder and represented humans via the **parameterized body model** (SMPL).
- Implemented a Temporal **Diffusion Transformer** (DiT-TS) backbone utilizing AdaLN-Zero modulation, dynamically **adapting to varying diffusion stages**.
- Introduced kinematic constraints (velocity/acceleration) to enforce physical plausibility, **enhancing temporal coherence** by up to 24.6%.

HOI-M3: Capture Multiple Humans and Objects Interaction within Contextual Environment

Third Author, Frontier Science Research Base on Intelligent Human-Machine Collaboration and Interaction

Shanghai, China

Aug 2023 - Mar 2024

- **Accepted by CVPR 2024 (Highlight - Top 3% of all submissions)**. (arXiv:2404.00299v2 [cs.CV])
- Co-created the **first large-scale MPMO interaction dataset** comprising 181M+ frames captured via synchronized RGB and object-mounted IMUs.
- Engineered a **computer vision interactive annotation pipeline** with SAM, generating pixel-accurate initial masks to bootstrap tracking across 42 views.
- Constructed 3D human ground-truth by implementing multi-view 2D keypoint **detection, trajectory triangulation**, and SMPL parameter **optimization**.
- Benchmarked SOTA **monocular 3D reconstruction models** by adapting to multi-entity scenarios, and engineered **3D visualization pipelines with Blender**.

3D Character Generation Using ControlNet and LoRA

Team Member, University of California, Berkeley

Berkeley, CA, USA

Nov 2023 - Dec 2023

- **Fine-tuned Stable Diffusion via LoRA** to enforce character identity consistency across generations, reducing FID by 15%.
- Integrated ControlNet with 3D human pose solutions to enforce **spatial consistency**, successfully **mitigating multi-view geometric artifacts**.
- Engineered the diffusion-based **Depth Control pipeline**, projecting depth priors via camera extrinsics to regulate body proportions for Gaussian Splatting.

Skills

Programming Python (PyTorch, NumPy, pandas, OpenCV, Gradio, vLLM), SQL

Miscellaneous Linux, LaTeX, Git, Docker, Blender